

Assignment 2

Question 1:

Scenario: A researcher uses linear regression to predict whether a tumor is malignant (1) or

benign (0). The predictions are all fractional values between 0.2 and 0.8.

Questions:

1. Why is linear regression not suitable for this problem?

Linear regression is not suitable because it's designed to predict continuous values, not fixed categories like malignant(1) or benign(0). The main issue is that a linear regression model draws a straight line. This line can and often will predict values outside the 0 to 1 range. These values are meaningless for a binary yes/no problem since we wouldn't be able to categorize them into class 0 and class 1. We need an output that can be interpreted as a probability, which linear regression doesn't provide.

2. What does logistic regression do differently?

Instead of just outputting the result of a straight line, logistic regression does an extra step. It takes the linear regression style calculation and passes it through a S shaped function called the sigmoid function. This function "squishes" the output, forcing any value (no matter how large or small) to fit neatly between 0 and 1. This is perfect for classification because the output can now be treated as a probability.

3. What does the sigmoid function represent in this context?

In this context, the output of the sigmoid function represents the estimated probability that the tumor belongs to the positive class (in this case, malignant or 1).

Eg: if the model outputs 0.7, it is 70% confident that the tumor is malignant. If it outputs 0.2, it is 20% confident it's malignant (and therefore 80% confident it's benign).

Question 2:

Scenario: A company uses kNN for classifying customer segments. When new data is added,

predictions suddenly change drastically.

Questions:

1. Why might kNN behave this way?

knn doesn't build a general model or rule from the data. Instead, it stores all the training data and uses it directly for every prediction. To classify a new point, it just looks at the k data points that are closest to it. When new data is added, the neighbourhood around any given point can change completely. A data point that was previously surrounded by one segment might now be closest to the new data, causing

its prediction to flip. This makes knn very sensitive to the exact data points it has stored.

2. What hyperparameter could stabilize its predictions?

The main hyperparameter to stabilize knn is k , the number of neighbours it checks. A very small k (like 1 or 3) is highly unstable because the prediction can be swayed by a single, noisy data point. Increasing k makes the model more stable. By checking more neighbours (eg $k=15$), the model has to get a broader consensus or majority vote, which averages out the influence of any single point and makes the predictions less volatile.

3. How does distance metric choice affect results?

The distance metric is the ruler that knn uses to measure closeness. The choice of metric can completely change what the model considers a neighbour.

The most common metric is Euclidean distance (a straight line). However, other metrics like Manhattan distance (which measures along grid lines like city blocks) might be better if your features have different scales or aren't naturally related in a straight line way. Changing the metric changes the shape of the decision boundaries and can significantly alter which data points are grouped together.

Question 3:

Scenario: A classifier gives high accuracy but fails to detect rare fraud cases.

Questions:

1. Why might accuracy be misleading here?

Accuracy is misleading because the dataset is imbalanced. Fraud is rare, so the not fraud class (the majority class) makes up almost all the data.

A lazy or dumb model can achieve very high accuracy by simply predicting not fraud every single time. If 99% of transactions are not fraud, this lazy model would be 99% accurate, which sounds great. However, it has completely failed at its real job it has zero ability to find the 1% of cases that are actually fraud.

2. Which other metric should be used?

Instead of accuracy, you should use metrics that focus on the rare (positive) class. The most important one here is Recall.

Recall answers the critical question: "Of all the actual fraud cases that happened, what percentage did our model correctly catch?" In this scenario, you want a very high recall.

You should also look at Precision which answers: "Of all the times the model alerted us to fraud, what percentage were actually fraud?" (This helps you know if you're wasting investigators time with false alarms).

3. How do ROC and AUC help understand model performance?

They help you see the model's performance beyond a single yes/no prediction.

ROC Curve: This is a graph that shows you the trade-off between the model's ability to catch true positives (like fraud) and how often it creates false positives (flagging

non-fraud as fraud). A good model's curve will bend up towards the top left corner (high true positives, low false positives).

AUC: This is a single number that summarizes the entire ROC curve. It tells you how good the model is at distinguishing between a fraud case and a non-fraud case. An AUC of 0.5 is no better than random guessing and an AUC of 1.0 is a perfect classifier. It's a much more reliable measure than accuracy for imbalanced problems.